

Inequalities of Polynomials

Basic skills

Q1. a) $4x^2 - 22x + 24 = 0$

$(2x - 8)(2x - 3) = 0$

$x = 4$

$x = \frac{3}{2}$

↘ factorise

↘ solve linear factors

b) $5x^2 + 6x - 12 = 0$

↘ formula

$$x = \frac{-6 \pm \sqrt{36 + 240}}{2 \times 5} = \frac{-3 \pm \sqrt{69}}{5}$$

Q2. The remainder of $P(x)$ on division by $ax - b$ is $P(\frac{b}{a})$

so $ax - b$ is a factor if and only if

$$P(\frac{b}{a}) = 0$$

Q3. Need to Perform factorising (or division)

$$3x^3 + x^2 - 12x + 4$$

$$= (x-2)(3x^2 + 5x - 2)$$

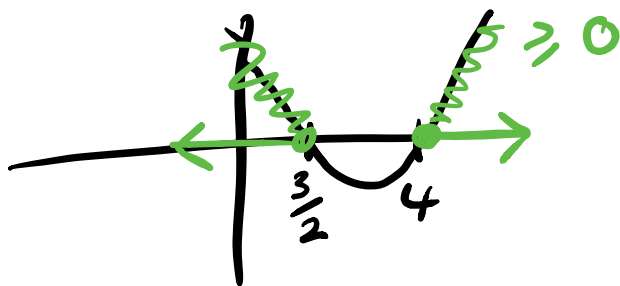
$\swarrow \quad \searrow \quad \swarrow \quad \searrow$
 $-6x^2 \quad -10x \quad \text{check } \checkmark$

$$= (x-2)(3x-1)(x+2)$$

so roots $x = 2, \frac{1}{3}, -2$

Q4. a) $4x^2 - 22x + 24 \geq 0$

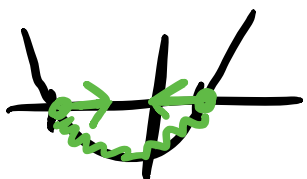
roots $x = \frac{3}{2}, 4$ by Q1 a)



so $x \geq 4$ or $x \leq \frac{3}{2}$

b) $2x^2 + 2x - 5 < 0$

$2x^2 + 2x - 5 = 0$ so $x = \frac{-2 \pm \sqrt{4 + 40}}{2 \times 2}$



$= \frac{-1 \pm \sqrt{11}}{2}$

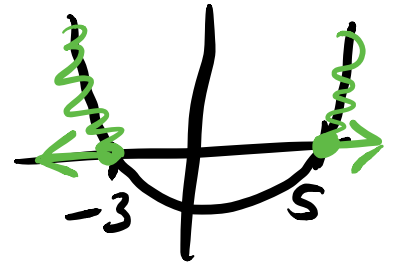
$$\text{So } \frac{-1-\sqrt{11}}{2} < x < \frac{-1+\sqrt{11}}{2}$$

c) $x^2 - 5 \geq 3x + 5$

$$x^2 - 3x - 10 \geq 0$$

$$(x - 5)(x + 3) \geq 0$$

$$x = 5, -3 \text{ roots}$$



$$x \geq 5 \text{ or } x \leq -3$$

Q5.

$$0 \leq x^2 - 2x - 3 \leq 5$$

$$x^2 - 2x - 3 = 0$$

$$(x - 3)(x + 1) = 0$$

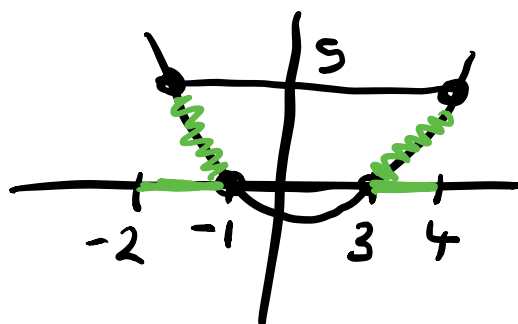
$$x = 3, -1$$

$$x^2 - 2x - 3 = 5$$

$$x^2 - 2x - 8 = 0$$

$$(x - 4)(x + 2) = 0$$

$$x = 4, -2$$



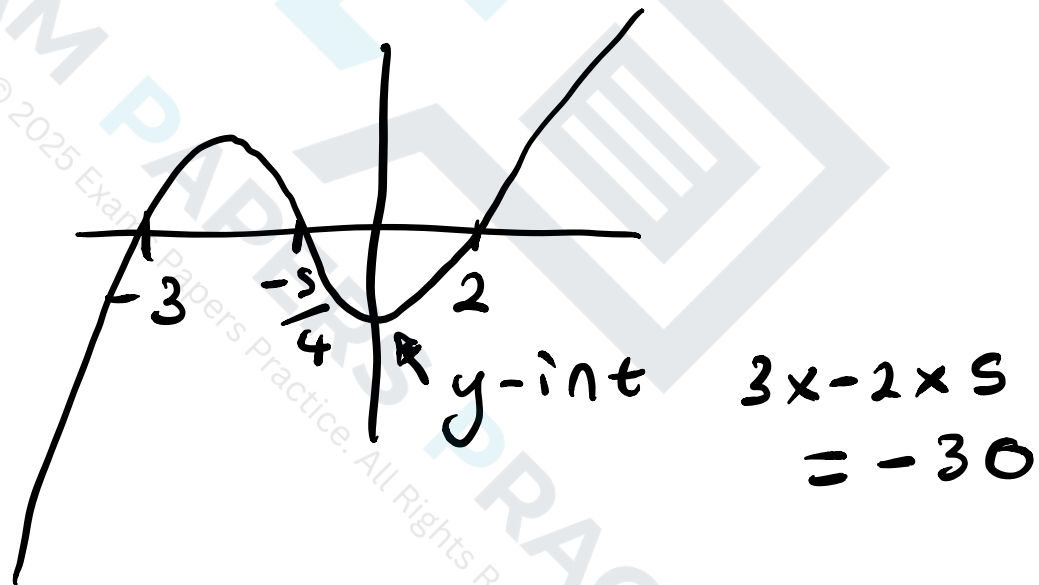
So $-2 \leq x \leq -1$ or
 $3 \leq x \leq 4$

Competency skills

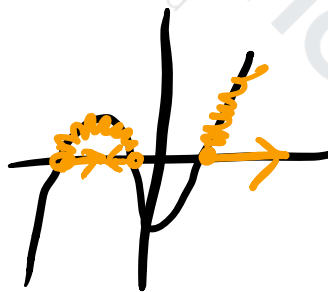
cubic (+ve)

Q6. a) $y = (x+3)(x-2)(4x+5)$

roots $x = -3, 2, -\frac{5}{4}$



b) for $y > 0$



$-3 < x < -\frac{5}{4}$ or $x > 2$

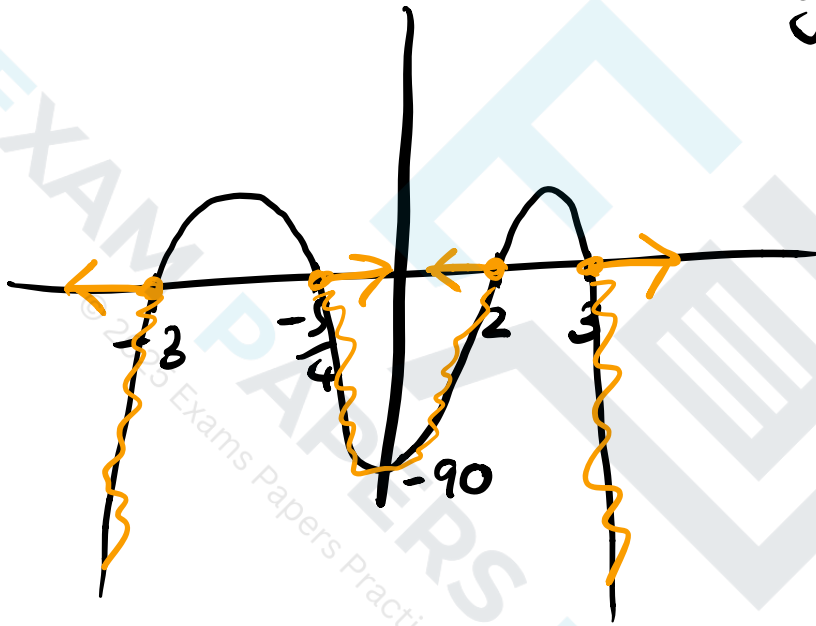
Q7. $(x+3)(x-2)(4x+5)(3-x) \leq 0$

↑ quartic (-ve)

roots $x = -3, 2, -\frac{5}{4}, 3$



y-int: $3 \times -2 \times 5 \times 3 = -90$



so $x \leq -3$ or $-\frac{5}{4} \leq x \leq 2$ or $x \geq 3$

Q8. $6x^4 - 5x^2 + 6x \geq 13x^2 - 6x + 4$

+ve quartic
W → $6x^4 - 13x^2 - 5x^2 + 12x - 4 \geq 0$

use factor theorem to 'guess' roots

$x = 1 : 6 - 13 - 5 + 12 - 4 = -4 \neq 0 \times$

$x = -1 : 6 + 13 - 5 - 12 - 4 = 0 \checkmark$

so -1 is a root

$$x=2: 6 \times 16 - 13 \times 8 - 5 \times 4 + 12 \times 2 - 4$$

$$= 0 \checkmark$$

so 2 is a root

so factors $x+1$, $x-2$

Two roots
is
enough to
get others by
factorising

$$(x+1)(x-2)(\sim)$$

$$(x^2 - x - 2)(\sim)$$

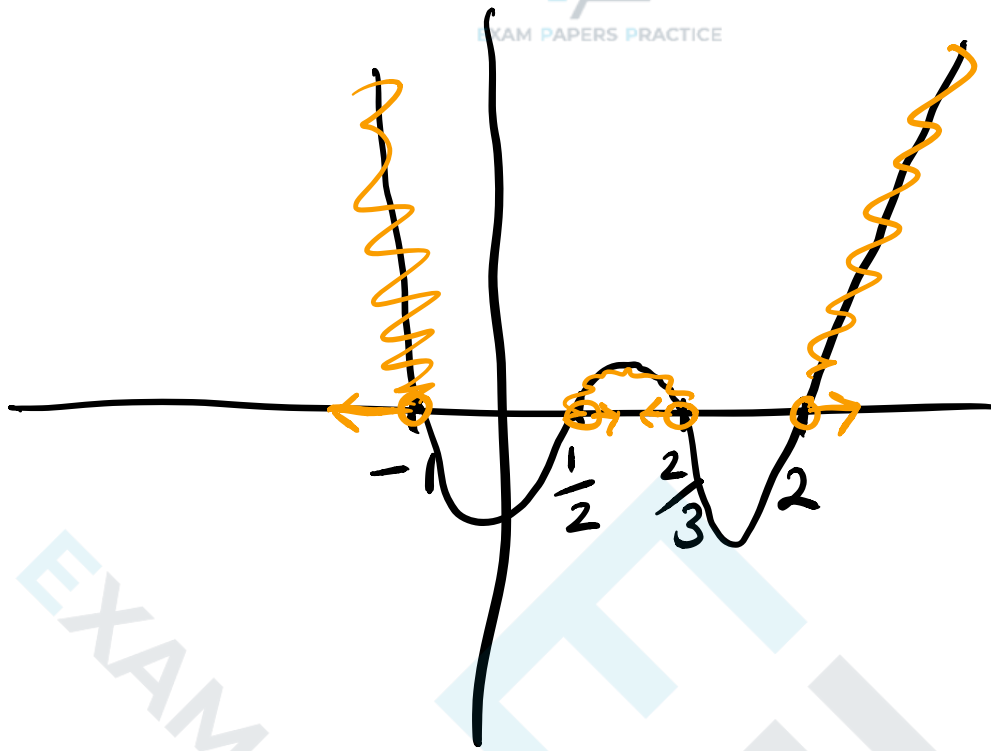
For remainder $\uparrow$

$$(x^2 - x - 2)(6x^2 - 7x + 2)$$

$$\downarrow \quad \uparrow \quad \downarrow \quad \uparrow \quad \downarrow$$
$$-6x^3 \quad +7x^2 \quad \text{check } \checkmark$$

$$(x+1)(x-2)(3x-2)(2x-1)$$

$$\text{roots } x = -1, 2, \frac{2}{3}, \frac{1}{2}$$



So $x < -1$ or $\frac{1}{2} < x < \frac{2}{3}$ or $x > 2$

Q9. a)

$$x^2 - 7 \geq \frac{6}{x} \rightarrow x^4 - 7x^2 \geq 6x$$

$\times x^2$

$$\text{so } x^4 - 7x^2 - 6x \geq 0$$

b) introduced solution $x=0$ as
multiplied by x^2 ← note. $x^2 \geq 0$ so
no need to
consider sign change

c) ^{roots} $x(x^3 - 7x - 6) = 0$

Factor theorem $x = -1$

gives $-1 + 7 - 6 = 0 \checkmark$

so $x - (-1) = x + 1$

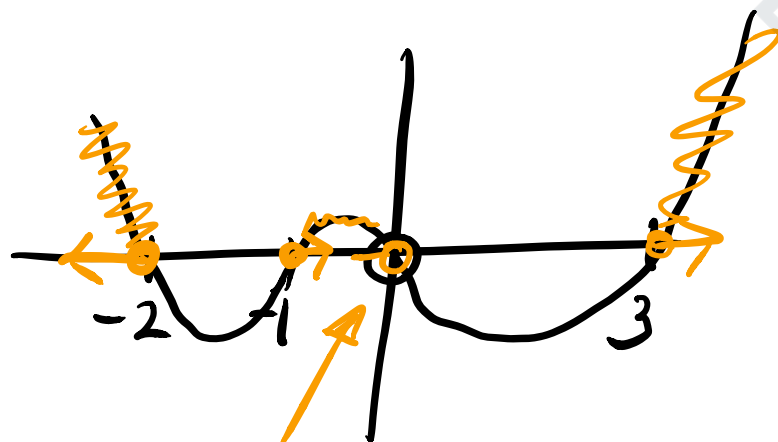
$$x(x+1)(x^2 - x - 6) = 0$$

$\downarrow \quad \nearrow \quad \downarrow \quad \nearrow \quad \downarrow$
 $x^2 \quad -x \quad \text{check } \checkmark$

$$x(x+1)(x-3)(x+2) = 0$$

$$x = 0, -1, 3, -2$$

now $x = 0$ is not a solution
as it was introduced



not = 0

So $x \leq -2$ or $-1 \leq x < 0$
or $x \geq 3$

d) Because $x \geq 0$ which may be negative, which would require a sign change

we would solve two inequalities

$$x^3 - 7x - 6 \geq 0$$

and

$$x^3 - 7x - 6 \leq 0$$

$$x > 0$$

$$x < 0$$

Advanced skills

Q8. a) $x^2 - x - 4 = x - x^2$

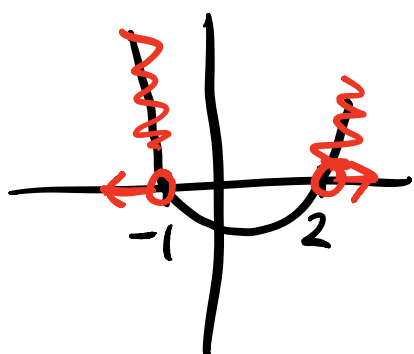
$$2x^2 - 2x - 4 = 0$$

$$x^2 - x - 2 = 0$$

$$(x+1)(x-2) = 0$$

$$x = -1, 2$$

b) so $x^2 - x - 2 > 0$



$x > 2$ or $x < -1$

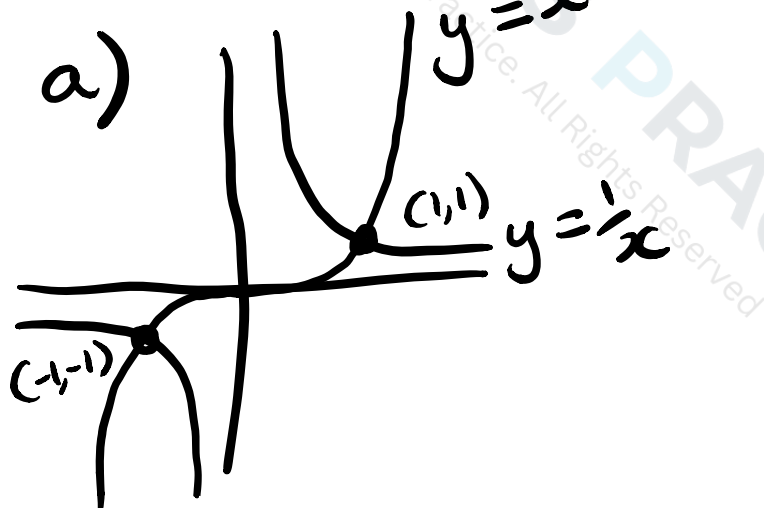
$2x - 5 \leq 0$



always $5 \leq \frac{5}{2}$,
so no need for
adjustment

so combining: $2 < x \leq \frac{5}{2}$ or $x < -1$

Q9. a)

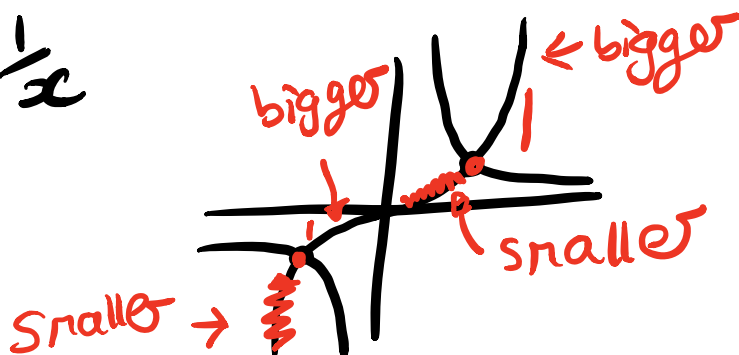


$x^5 = \frac{1}{x}$

$x^6 = 1$

$x = \pm 1$

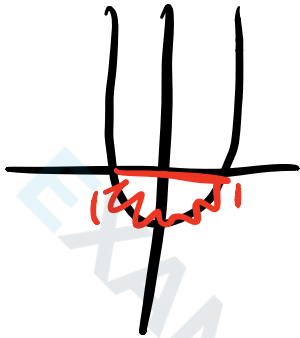
b) $x^5 \leq \frac{1}{x}$



so

$$x \leq -1 \text{ or } 0 < x \leq 1$$

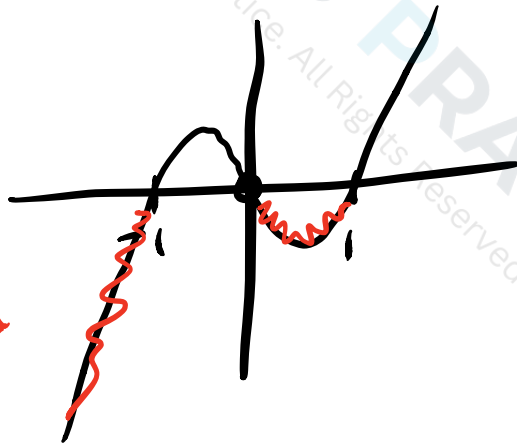
c) solving $x^6 \leq 1$ gives
roots $x^6 - 1 = 0$ $x = \pm 1$
 $-1 \leq x \leq 1$



Not same, misses a lot

solving $x^7 \leq x$ gives $x(x^6 - 1) \leq 0$
roots $x = 0, \pm 1$

may actually be, but no need to worry →



$$x \leq -1$$

or

$$0 < x \leq 1$$

- Issue with $x^6 \leq 1$ is assumes $x > 0$
to $x \cdot x$ in $x^3 \leq \frac{1}{x}$ missing issue with
 $x < 0$
so gets us $0 < x \leq 1$

- $x^7 \leq x$ needs $x=0$ removed if
 $x^5 \leq \frac{1}{x}$ as $x \times x^2$ giving $x=0$ sol
 so

$$x \leq -1 \text{ or } 0 < x \leq 1$$

Q10. $x+2 < \frac{1}{x-2}$

$\times (x-2)^2$ as always ≥ 0

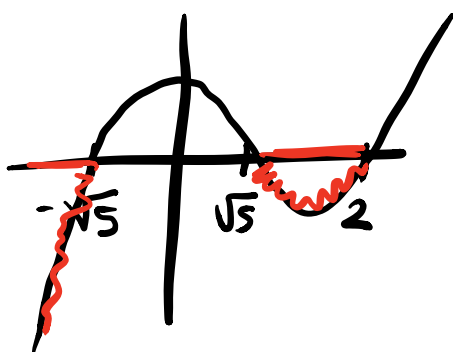
and remove $x=2$ sol.
if present

so $(x+2)(x-2)^2 < (x-2)$

$$(x-2)((x+2)(x-2) - 1) < 0$$

$$(x-2)(x^2 - 5) < 0$$

cubic roots $x = 2, \pm\sqrt{5}$



need $x \neq 2$

so
 $x < -\sqrt{5}$
 or

$$\sqrt{5} < x < 2$$

$x \neq 2$
 so
 no
 need to
 consider
 removal